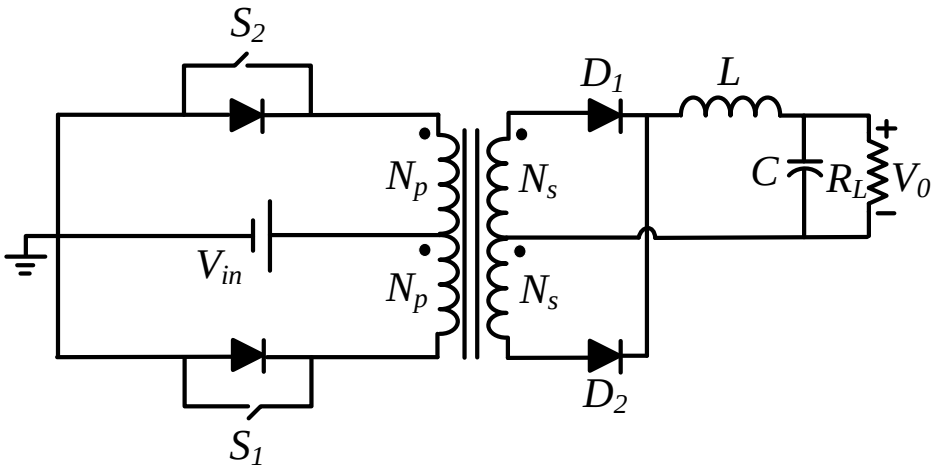
WEEK 7: ASSIGNMENT

1. 1. For push-pull converter shown below, what will be the output voltage, if $V_{in} = 50V$, $N_s = 5$, $N_p = 2$ and duty ratio (D) is 0.4.



Answer: 50

Range can be between 49.5 to 50.5

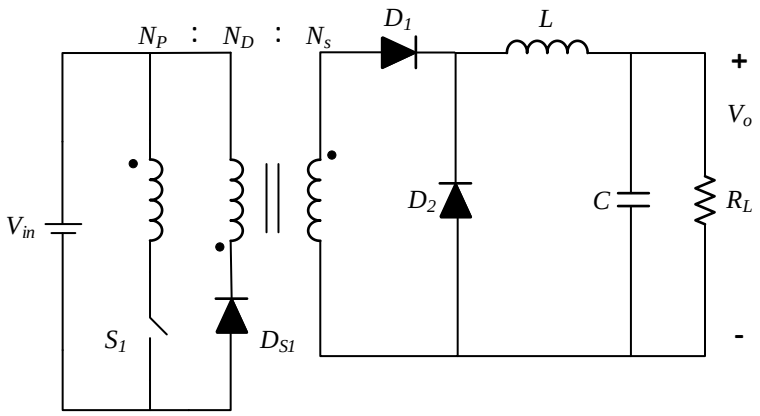
2. In dual switch forward converter, the voltage stress on the switch is equal to
A. The input voltage
B. Twice the input voltage
C. The input voltage plus the voltage spike due to leakage inductance of the primary winding
D. Twice the input voltage plus the voltage spike due to leakage inductance of the primary winding

Answer: A

3. In forward converter with demagnetizing winding, have the turns ratio as $N_p : N_s : N_D = 1 : 2 : 3$. Assume the $L_{lkg} = 0$. Then, what will be the voltage stress on the switch (S_1)? Symbols have the obvious meaning
A. V_{in}
B. $\frac{1}{3}V_{in}$
C. $\frac{2}{3}V_{in}$
D. $3V_{in}$
E. $4V_{in}$

Answer: C

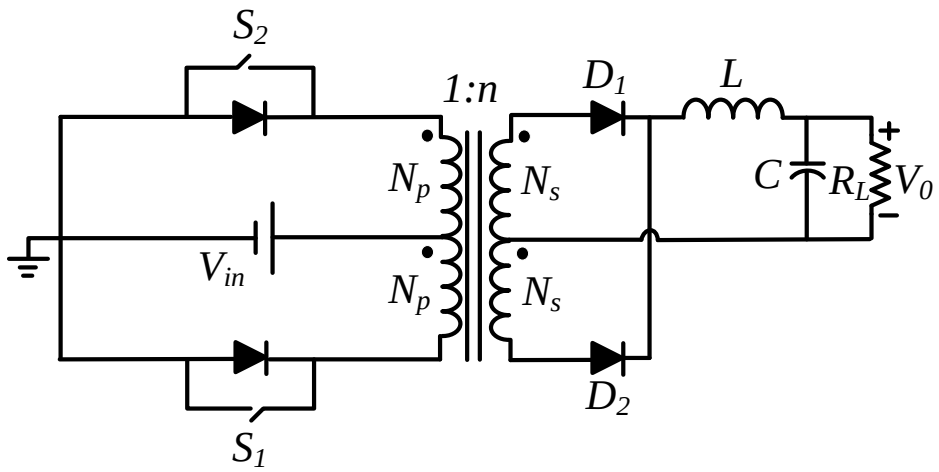
4. For the forward converter shown below, if the number of turns on demagnetizing winding (N_D) is three times the number of turns on the primary winding (N_P). Then what is the maximum time (μs) for which switch (S_1) can be turned on, if the switching frequency is kept at 10kHz. (Note: The answer should be rounded up to 2 decimal places)



Answer: 25

Range can be between 24 to 26

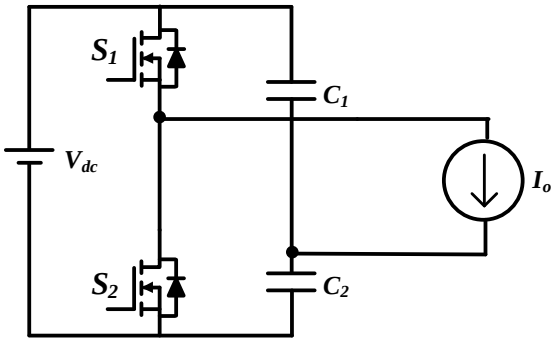
5. For the isolated DC-DC converter shown below, what will be the maximum voltage stress on the switches (S_1 and S_2), if $n = 2$



A. V_{in}
B. $2V_{in}$
C. $0.5V_{in}$
D. $3V_{in}$

Answer: B

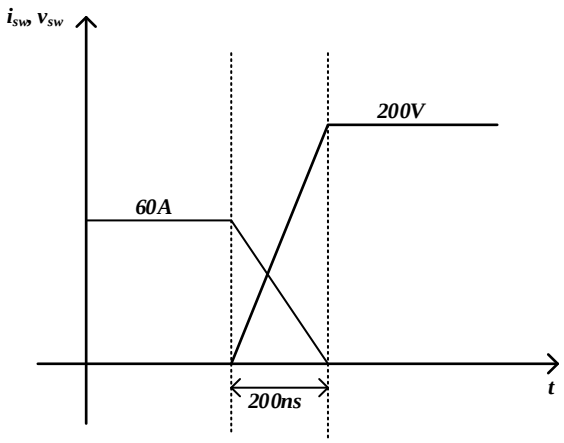
6. For the circuit shown below, which all switches will obtain the zero-voltage switching (assume sufficient deadtime is provided and necessary condition for zero-voltage switching is met).



A. S_1
B. S_2
C. Both S_1 and S_2

Answer: B

7. A power switch has following $i_{sw}(t)$ and $v_{sw}(t)$ profile when turned-off. Find the energy loss (in μJ) during turn-off.



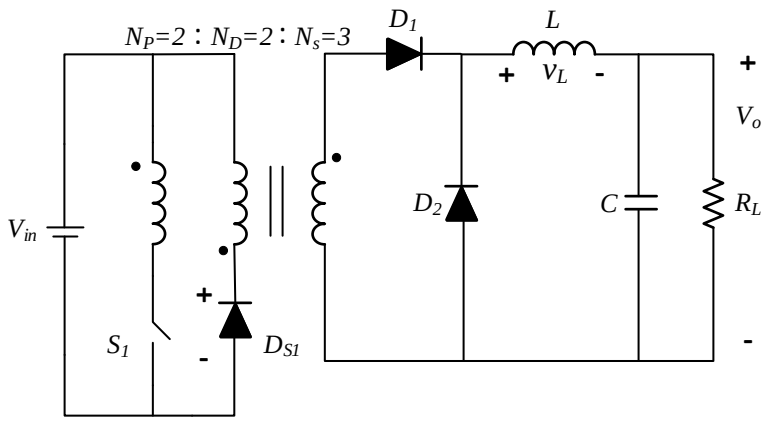
Answer: 400

Range can be between 399 to 401

8. Which of the following statement/statements are true about the isolated DC-DC converter?
A. In full- bridge converter, the flux density swings between 0 to B_{max}
B. In full- bridge converter, the flux density swings between $-B_{max}$ to B_{max}
C. In forward converter, the flux density swings between 0 to B_{max}
D. In forward converter, the flux density swings between $-B_{max}$ to B_{max}

Answer: B, C

9. For the forward converter shown below, what will be the output voltage if the $V_{in} = 100\ V$, when the switch S_1 is switching with the frequency of 20 kHz and is turned ON for 20 µs. The answer should be rounded up to 2 decimal places.



Answer: 60

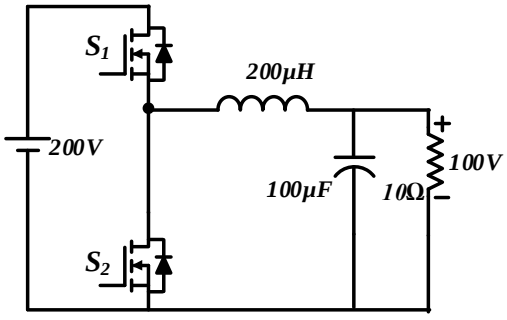
Range: 59 to 61

10. A Full bridge converter supplying a 500W, 100V load from an input voltage of 400V. What will be the value of output inductance (in mH) to be kept to limit the inductor current ripple (peak to peak) within 5% of full load current? The transformer used has 4 numbers of primary winding and 2 numbers of secondary winding. The converter is switching at a frequency of 20 kHz. The answer should be rounded up to 2 decimal places.

Answer: 5

Range: 4 to 6

11. For the circuit shown below, if the S_2 switch achieve the ZVS (assume the negligible ripple on the output current coming out of half bridge), then what will be the $\frac{dv}{dt}$ (in $\frac{V}{ns}$) at the output of the half-bridge when switch S_1 turns-off . If the C_{oss} of the switches S_1 and S_2 are 1nF. The answer should be rounded up to 2 decimal places.



Answer: 5

Range: 4 to 6

12. Which of the following statement/statements is/are correct about the soft switching in conventional half-bridge converter. Symbols have obvious meaning

A. The ZVS condition is needed to make zero switching losses during turned-on conditions
B. The ZVS condition is needed to make zero switching losses during turned-off conditions
C. The ZCS condition is needed to make zero switching losses during turned-on conditions
D. The ZCS condition is needed to make zero switching losses during turned-off conditions

Answer: A, D

Answer Key:

1. 49.5 to 50.5	2. A
3. C	4. 24 to 26
5. B	6. B
7. 399 to 401	8. B, C
9. 59 to 61	10. 4 to 6
11. 4 to 6	12. A, D